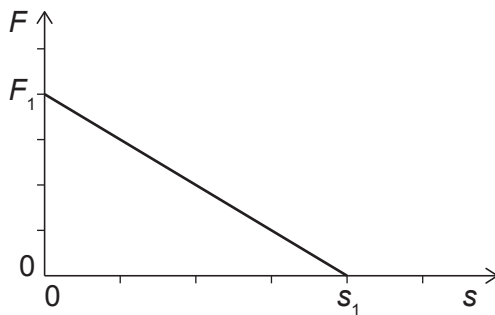


PAPER 1 / QUESTION 96

4 November 2021 / TZ0 / HL Paper 1 / original question 4 / 8821-6501

96. A net force F acts on an object of mass m that is initially at rest. The object moves in a straight line. The variation of F with the distance s is shown.



What is the speed of the object at the distance s_1 ?

- A. $\sqrt{\frac{F_1 s_1}{2m}}$
- B. $\sqrt{\frac{F_1 s_1}{m}}$
- C. $\sqrt{\frac{2F_1 s_1}{m}}$
- D. $\sqrt{\frac{4F_1 s_1}{m}}$